Toronto Metropolitan University

Metropolitan Undergraduate Engineering Society

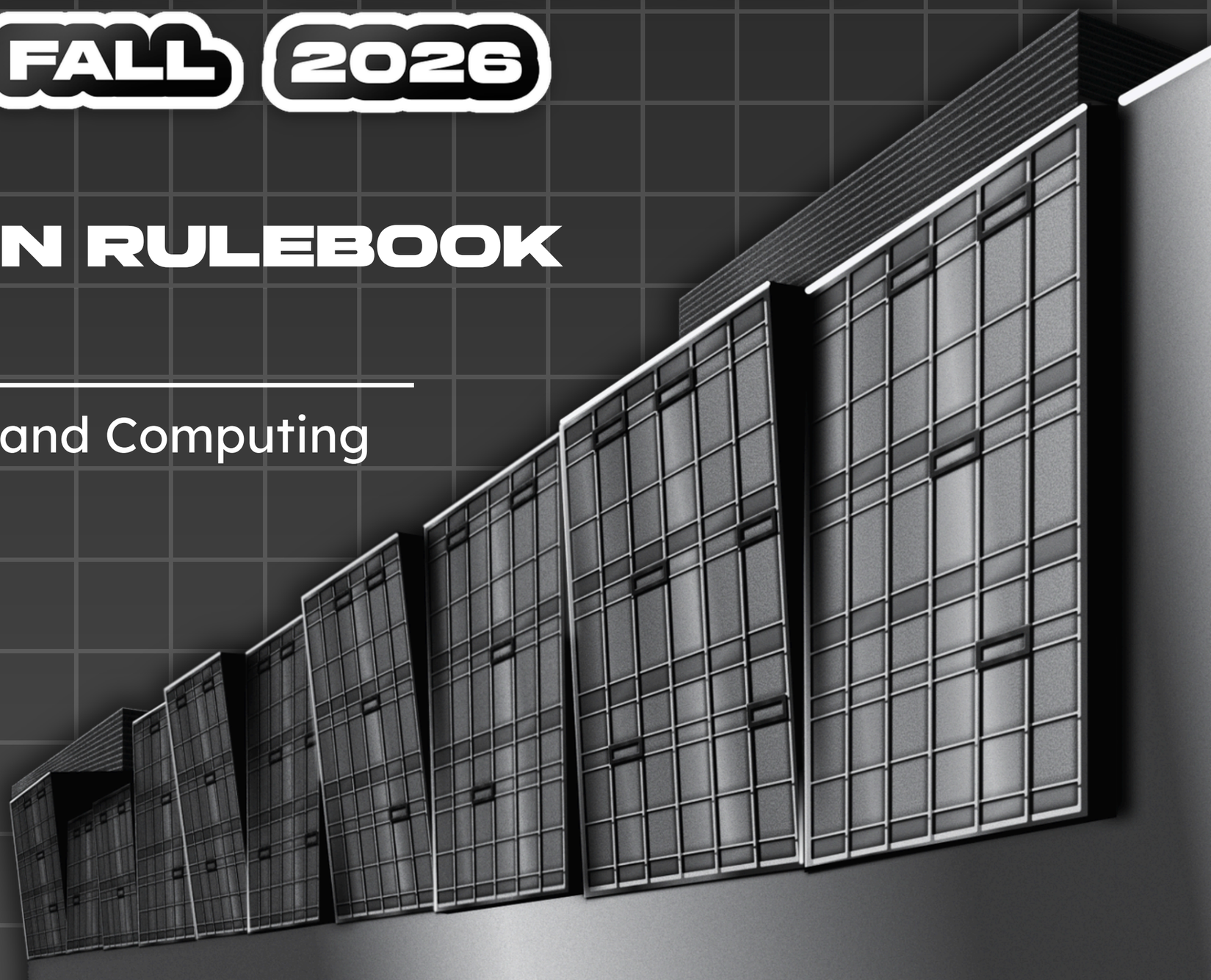
TORONTO METROPOLITAN UNIVERSITY ENGINEERING COMPETITION

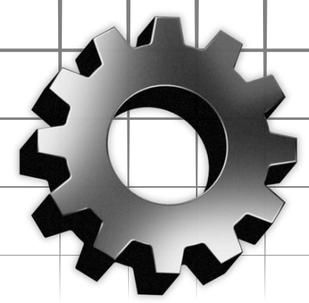
FALL 2026

JUNIOR DESIGN RULEBOOK

NOVEMBER 7 & 8, 2026

George Vari Engineering and Computing Centre





This competition challenges junior engineering students to design and build a prototype to address a technical problem. The Junior Team Design category is similar to the Senior Team Design competition, but emphasis is placed on prototype functionality rather than design theory.

1.1. TEAM COMPOSITION

Each Junior Design team must consist of a minimum of three (3) and a maximum of four (4) competitors. The team must be entirely composed of students who must not have completed more than half of the required credits for their accredited engineering program.

1.2. LANGUAGE

The presentation and any optional presentation materials can be done in either English or French but must be consistent in language.

1.3. RESOURCES

1.3.1. Facilities Required

- One (1) amphitheater (presentation of problem and team presentations)
- One (1) large workroom, subdivided with partitions for each team or one (1) separate workroom per team
- One (1) centralized competition presentation/testing space (location will depend on the constraints of the design problem)

1.3.2. Personnel Required

1.3.2.1. Judges

A minimum of three (3) judges (and in any excess, an odd number of judges) are required to assess the problem-solving abilities, proposed solution, communication skills and team dynamics of the competitors. Judges should possess sufficient technical knowledge or relevant engineering experience to evaluate prototype functionality, design decisions, testing performance, and engineering reasoning. Effective communication, teamwork, and presentation skills will also be considered during assessment.

1.3.2.2. Competition Leads

The Competition Leads are responsible, along with the VP Competitions, for the entire design and implementation of the competition. The Competition Leads must present the design problem at the beginning of the competition and answer any questions. Only the Competition Leads may answer questions during the design period. The Competition Leads will also answer any questions the judges may have during the competition.

1.3.2.3. Official Timekeeper

The Official Timekeeper is responsible for enforcing time limits during the solution development period and the presentations.

1.3.3. Equipment Supplied by MEC

The following equipment will be made available to all competitors during the competition time period:

1.3.3.1. During the design phase

- A design area with at least one (1) table, four (4) chairs and at least one (1) of a whiteboard, chart paper or blackboard
- Paper and pencils/pens for writing
- Any materials and/or tools specific to the design problem
- Power source if powered tools supplied
- Access to an MEC Item Shop where teams may use their allocated Engineering Dollars to purchase approved materials and components. Available items, prices, purchase limits, and inventory restrictions will be communicated by the Competition Organizers.

1.3.3.2. During the presentation

- One (1) table
- One (1) projector
- Whiteboard(s) or blackboard(s)

1.3.4. Allowed External Resources

The competitors are allowed to bring the following equipment with them:

- Any textbooks, course notes, or other reference material
- Digital camera
- Each team member is allowed one device*
- Any additional tools, equipment and/or software as deemed fit by the competition director so long as it is reasonable to expect all participants to have access to said tools, equipment and/or software

Note: Devices may include a computer, iPad, tablet, or other electronics of similar nature. A competitor may have multiple devices present but is only permitted to use one (1) at a time. The use of the internet to aid in the design process is prohibited.

Competitors may prepare blank presentation templates before the competition, provided the templates do not contain pre-developed technical solutions or competition-specific design content. All presentation materials must comply with the requirements communicated by the Competition Organizers.

1.4. PROCEDURE / TIMELINE

1.4.1. Pre-Competition

Prior to the competition, official rules, schedules, and relevant competition information will be communicated to competitors through authorized MEC communication channels.

1.4.2. Competition

1.4.2.1. Presentation of Problem

The problem will be presented to all competitors and judges at the beginning of the competition. The Competition Leads will provide detailed explanations of what is expected from the competitors, both orally and in writing.

1.4.2.2. Question Period

Competitors then have fifteen (15) minutes to ask the Competition Leads any questions they may have. Only the time used to ask the questions (not the answers) should be counted. The answers will be provided orally and recorded in written form.

1.4.2.3. Development of the Solution

Teams will have eight (8) hours to develop their solutions, construct their prototypes, complete permitted practice testing, and prepare the required deliverables. Official testing will take place according to the schedule established by the Competition Organizers. The Competition Organizers will announce the official start time, submission deadline, and testing schedule during the competition briefing. All deliverables must be submitted to the Competition Leads before the allotted time expires. Competitors may finish before the end of the allotted time.

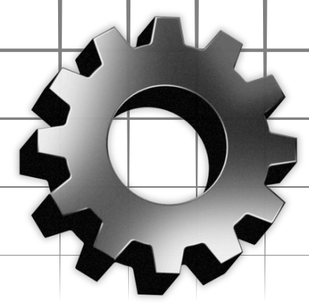
1.4.2.4. Rest Period

Competitors must be allowed a minimum of one (1) hour to rest before the presentation phase starts.

1.4.2.5. Presentation and Tests

Competitors have ten (10) minutes to present their designs. All team members must be present and participate in the presentation. The original design, its rationale, a description of the design process, and a critique of the prototype are required presentation components. The judges then have a maximum of five (5) minutes to ask questions.

The completed prototype must be available for inspection and evaluation by the judges. Official prototype testing may be conducted separately from the presentation according to the schedule announced by the Competition Organizers. Each team will participate in the official prototype testing procedure established by the Competition Organizers.



Testing may be scheduled separately from the presentation phase to ensure that all teams are evaluated fairly and efficiently. The number of permitted attempts, testing duration, performance requirements, and applicable deductions will be communicated during the competition briefing. All teams will be assessed using the same standardized testing procedures.

1.4.3. Timekeeping

The following rules will be adhered to with respect to timekeeping.

1.4.3.1. During the design phase

- Time for the solution development period is started when all the teams have reached their work areas
- The remaining time for solution development must be announced two (2) hours, one (1) hour, thirty (30) minutes, and ten (10) minutes before the end of the allotted time

1.4.3.2. During the presentation

- Time is halted when a judge asks or answers a question while the problem is being presented
- Only the time used to ask questions during the question period is counted, not the time used to answer questions
- The remaining presentation time must be indicated to the competitors two (2) minutes and one (1) minute before the end of the allotted time
- A visual countdown must be given during the last thirty (30) seconds of the presentation to the judges
- After the ten (10) minute presentation time, the competitors will have a fifteen (15) second grace period to conclude their presentation before they are cut off to begin the question period

1.4.4. Presentation Order

- Presentation order shall be determined randomly
- Official prototype testing may follow a separate schedule from the presentation order. Testing assignments and submission deadlines will be communicated by the Competition Organizers.
- Presentation order shall be announced thirty (30) minutes before the presentations commence. All teams are required to be present at this announcement
- Teams are not allowed to switch places in the presentation order
- Teams may not be in the audience for presentations of their competition until after they have completed their own presentation. Competition Leads will enforce this restriction consistently for all teams.

1.4.5. Deliverables

Each team must complete a functional physical prototype and submit all required presentation materials before the official submission deadline. The prototype will be evaluated through standardized testing procedures established by the Competition Organizers. Teams must also present and defend their engineering design decisions before a judging panel. Detailed deliverable requirements, submission formats, and deadlines will be provided in the official competition package.

1.4.6. Response to Questions

- Only the Competition Leads may answer questions during the problem presentation and design phases. Volunteers and judges may not answer questions regarding rules and procedures at any time
- During the presentation of the problem, the Competition Leads shall answer orally and write down the answers provided
- During the design phase, only questions related to deliverable content shall be answered. No answers shall be provided in response to questions about the problem that might lead to the development of a new approach or which might invalidate a solution
- During the design phase, answers to questions shall be provided in writing to all teams at the same time. Fifteen (15) minutes may be allowed to elapse for a significant number of questions accumulated before responses are provided. A copy of the responses must be provided to the judges prior to the presentations

1.5. ASSESSMENT AND JUDGING

- The panel must have an odd number of judges
- The panel must have at least three (3) judges
- The assessment shall be carried out in the audience's presence
- Feedback forms shall be provided to each team following the announcement of winners but prior to the end of MEC

1.6. SOLUTION AND RUBRIC

- The winning solution will not necessarily be the most technically effective solution, but the solution that has the most real-world applicability and forethought.
- All solutions will be evaluated based on their functionality, practicality, engineering design, compliance with competition constraints, and overall performance. Designs that fail to satisfy the competition requirements will receive scores reflecting their performance under the official judging criteria.

1.6.1 Judging Matrix

Category	Judging Criteria	Points
Design Quality	Practicality, sustainability, scalability, and ease of manufacturing	/15
	Structural integrity, functionality, and reliability of the proposed design	/10
	Material and budget efficiency	/10
	Compliance with competition constraints and safety requirements	/5
Design Quality Subtotal		/40
Testing	Ice preservation measured as the percentage of solid ice remaining	/20
	Obstacle-course completion, stability, and towing reliability	/10
Testing Subtotal		/30
Presentation	Explanation of the design process, materials, and engineering decisions	/15
	Justification of costs and material choices	/5
	Effective use of visual aids	/5
	Responses to judges' questions	/5
Presentation Subtotal		/30
TOTAL		/100

1.6.2 Competition Penalties

Violation	Points Deducted
Plagiarism	–50
Insufficient citations	–50
Documents received after the official deadline	–50
Absent team member during a required competition activity	–25